

Solved Problem(FLSM)

Problem#1

An organization has been assigned the IP block **192.168.10.0/24**.

They want to divide this network into **4 equal subnets** using **FLSM**.

Tasks:

1. Determine the new subnet mask.
2. List the **network address**, **first usable IP**, **last usable IP**, and **broadcast address** of each subnet.
3. How many usable hosts are available in each subnet?

Solution:

The given IP block 192.168.10.0/24.

The prefix length is 24. No. of host bits: $32-24=8$. Hence, there are $2^8 = 256$ addresses in this block.

a)

Since this is FLSM, all subnet will have the same subnet mask. To find the subnet mask, write 24 ones and 8 zeros as follows:

11111111.11111111.11111111.00000000=255.255.255.0

b)

Since this is FLSM, we have to allocate these 256 IPs among the 4 subnets equally (equally because this is FLSM).

Steps:

1. The number of IPs per network is $256/4=64$.
2. No. of host bits per subnet= $\lceil \log_2(64) \rceil = 6$
3. No. of net bits per subnet= $32-6=26$
4. The given IP block can be represented in binary as .

11000000.10101000.00001010.00000000



Prefix length of the IP block (24)—Chosen from the left most bits

No. of host bits/subnet (6)—Chosen from the right most bit

The bits in between the net bits and host bits/subnet are subnet ID bits.

The network IP address of each subnet can be found by incrementing these subnet bits.

Network IP of the first subnet:

11000000.10101000.00001010.00000000=192.168.10.0

Network IP of the second subnet:

It can be obtained by incrementing 00 to 01

11000000.10101000.00001010.01000000=192.168.10.64

Network IP of the third subnet:

It can be obtained by incrementing 01 to 10

11000000.10101000.00001010.10000000=192.168.10.128

Network IP of the Fourth subnet:

It can be obtained by incrementing 10 to 11

11000000.10101000.00001010.11000000=192.168.10.192

Subnet no.	Network Address	First Usable IP	Last Usable IP	Broadcast IP
1	192.168.10.0	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

c) We allocate 64 IPs to each subnet. Of them 2 IPs are reserved for network and broadcast ip, hence, the number of host IPs=64-2=62.

Problem#2

You are given the network **172.16.0.0/19** for 4 departments with the following host requirements:

- Dept A: 700 hosts
- Dept B: 50 hosts
- Dept C: 50 hosts
- Dept D: 25 hosts

Find the network IP addresses of each network.

Solution:

1. Largest network A: 700
2. $H_2 = 700 + 2 = 702$
3. No. of host bits in subnets: $\lceil \log_2(700) \rceil = 10$
4. No. of net bits in subnets: $32 - 10 = 12$
5. No. of IPs to be allocate to each subnet: $2^{10} = 1024$

Network IP address of A

10101100.00010000.00000000.000000000000 Subnet ID bits

Network IP: 172.16.0.0

Network IP of B

10101100.00010000.00000001.000000000000 Subnet ID bits

172.16.4.0

Network IP of B

10101100.00010000.00000010.000000000000 Subnet ID bits

172.16.8.0

Network IP of B

10101100.00010000.00000011.000000000000 Subnet ID bits

172.16.12.0